

Time and Day-Based Peak Electricity Demand Forecasting: A Comparative Analysis of Machine Learning Models for Peak Cut and Decarbonization

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Abstract— This study aims to address peak power cuts and shifts in Japan by forecasting peak electricity demand using machine learning models. The focus is on a method that excludes meteorological data and utilizes only time-based features. This approach enables medium-term forecasts one month ahead, making it feasible for operational planning without reliance on weather forecasts. It also reduces computational requirements, avoiding the need for high-spec central processing units or graphical processing units. We evaluate and compare several models, including Random Forest, XGBoost, K-Nearest Neighbors, Decision Tree, and LightGBM, to determine their accuracy in providing reliable forecasts. The LightGBM model, in particular, demonstrated remarkable results, achieving forecasting accuracies of 60 in June 2024 and 74 in July 2024. This indicates that it is a highly accurate and computationally efficient model, significantly enhancing peak demand forecasting precision. This approach facilitates peak cut and peak shift strategies, which help avoid electricity shortages and achieve stable demand and supply. Furthermore, it contributes to decarbonization by eliminating unnecessary waste.

Keywords— *Peak Electricity Demand Forecasting; Machine Learning Models; Feature Engineering; Electricity Demand Peak Time*

I. INTRODUCTION

The electricity capacity market in Japan is a crucial mechanism for maintaining energy supply stability, requiring power companies and generators to guarantee a certain level of supply capacity. The system calculates capacity contribution fees based on peak electricity demand, which is directly linked to cost reduction [1]. Lower peak demand results in a smaller amount allocated to each company, reducing overall payments. Specifically, the Regional Grid Operators (known as "Regional Operators") set national supply capacity goals and establish contracts through auctions to meet these goals. The capacity contribution fees are calculated based on these contracts, using summer peak demand (July to September) and winter peak demand (December to February) with adjustments made according to the previous year's demand performance. Additionally, Demand Response (DR) programs are designed to reduce electricity demand during peak times, making accurate demand forecasting essential for the success of these programs [2]. Traditional forecasting models often include complex

factors such as weather data, which can affect the accuracy and practicality of predictions [3]. By eliminating dependence on weather data and focusing solely on time-based features, this new approach aims to provide a simpler and more broadly applicable method for energy management.

This study proposes a novel approach to forecasting peak electricity demand by using only time-based features and excluding weather data. This method enables medium-term forecasting, such as one-month-ahead predictions, facilitating more strategic operational planning. Furthermore, it does not rely on high-specification computational hardware, making it applicable to a wider range of energy management scenarios. Specifically, we evaluate the performance of several machine learning models—Random Forest [4], XGBoost [5], KNN [6], Decision Trees [7], and LightGBM [8]—to determine their effectiveness in predicting peak demand. Notably, the study highlights the achievements of the LightGBM model, focusing on its high accuracy and potential for enhancing peak cut and peak shift strategies, thereby avoiding electricity shortages, achieving stable demand and supply, and contributing to decarbonization by reducing unnecessary waste.

II. METHOD

A. Data Collection and Preprocessing

The dataset for this study was collected from Tokyo Electric Power Company (TEPCO) and covers the period from 2020 to 2024 [9]. The dataset includes hourly electricity usage data, which was converted from megawatts (MW) to kilowatts (kW) for consistency and ease of analysis. The data was split into training and testing datasets: the training data spans up to June 30, 2023, while the testing data extends from July 1, 2023, to September 30, 2024. This division allows for training the models on historical data and evaluating their performance on subsequent data. While some studies [10]–[12], such as Neumann et al. [10], use temperature data to improve prediction accuracy, this study focuses on predicting the peak hour of the day using minimal data necessary for mid- to long-term predictions. Therefore, only past demand data and calendar information were collected for the analysis, without including temperature data.

B. Enhanced Feature Engineering

In this study, several features were engineered to improve the accuracy and relevance of the predictions. These features aim to enhance the comprehensiveness and precision of the electricity demand forecasting model.

Year of Observation

- **Extraction Method:** The year was extracted from the dataset to capture annual variations.

- **Rationale:** Including the year as a feature allows the model to account for long-term trends and variations in electricity demand. Factors such as economic growth, changes in energy consumption patterns, and infrastructure developments over different years can influence demand. By incorporating the year, the model can consider these broader temporal shifts, potentially improving prediction accuracy. This approach helps the model reflect changes across a larger time scale and achieve more precise forecasts.

Holiday Flag

- **Extraction Method:** The holiday flag was created using the Japanese public holiday calendar to identify whether a specific date is a public holiday.

- **Rationale:** Public holidays often exhibit distinct electricity usage patterns compared to regular weekdays. For example, there may be reduced demand due to decreased business activity or increased demand due to special events. Including a holiday flag allows the model to capture these deviations more accurately in demand, leading to improved prediction accuracy during holiday periods. This feature not only enhances forecasting precision but also contributes to the effective implementation of DR programs.

These features, shown in Table 1, are critical elements in the electricity demand forecasting model, capturing complex temporal patterns, and significantly enhancing prediction accuracy. This setup assumes that Japan's demand is heavily influenced by the distinct differences between weekday and weekend lifestyles, largely due to the prevalence of full-time employment. Additionally, variations in factory operations between weekdays and weekends are considered, as they have a significant impact on overall electricity consumption patterns.

TABLE I. FEATURE SET

Feature	Description
Hour of the Day	Captures the diurnal patterns in electricity usage.
Day of the Week	Accounts for weekly usage patterns.
Month of the Year	Reflects seasonal variations in electricity demand.
Day of the Month	Identifies day-specific effects on demand.
Is Weekend	Binary feature for weekend vs. weekday
Year of Observation	Captures long-term yearly trends.
Holiday Flag	Identifies public holidays for special demand.

C. Model Selection

In this study, several machine learning models were evaluated to determine their effectiveness in predicting electricity demand, with each model chosen based on its ability to handle various aspects of the data and its suitability for the forecasting task. The Random Forest Regressor, an ensemble method, leverages multiple decision trees to model complex relationships and interactions within the data, effectively capturing intricate patterns in electricity demand due to its strength in handling non-linearities. XGBoost Regressor, known for its robustness and accuracy, is a gradient boosting technique that builds models iteratively to minimize prediction errors. It performs well on large datasets, with its regularization capabilities helping manage overfitting and improve prediction performance. The KNN method, a non-parametric approach, classifies data based on the similarity of data points. While intuitive and straightforward, making it useful for less complex relationships between features, its performance can be sensitive to the choice of distance metric and the number of neighbors. The Decision Tree Regressor splits data into subsets based on feature values, providing a clear and interpretable decision process, though it may suffer from overfitting if not properly tuned or pruned.

Finally, the LightGBM Regressor, a gradient boosting framework optimized for large datasets, is known for its high efficiency and speed, making it suitable for handling large-scale data and complex models with faster training and lower memory usage compared to other boosting algorithms. Each of these models was evaluated to understand their performance in predicting peak electricity demand, considering their strengths and limitations, with the choice of models reflecting a balance between accuracy, computational efficiency, and scalability, aimed at achieving the best possible forecasting results.

D. Model Training and Evaluation

The models were trained and evaluated using the enhanced feature set, with the training process utilizing data from the historical period from January 1, 2020, to December 31, 2023, while the testing process applied to the period starting from January 1, 2024. This approach ensures that the models are trained on past data and evaluated on future data, providing a realistic assessment of their predictive performance. The models were evaluated based on several metrics, including Mean Absolute Error (MAE), Mean Squared Error (MSE), and R^2 Score. These metrics offer insights into the accuracy of the predictions and the models' ability to explain the variance in the demand data. By incorporating the year and holiday features, the models were able to capture both long-term trends and specific day-to-day variations, leading to improved forecasting accuracy. This methodology outlines a comprehensive approach to forecasting peak electricity demand using temporal features alone, demonstrating how feature engineering can enhance predictive performance and practical utility in energy management.

TABLE II. COMPARISON OF MODEL PERFORMANCE

Model	MAE(10^6 kW)	MSE(10^{12} kW ²)	R ² Score
Random Forest	3.03	19.44	0.723
XGBoost	2.92	16.43	0.766
KNN	3.21	20.14	0.714
Decision Tree	3.58	28.21	0.599
LightGBM	2.71	13.77	0.804

III. RESULTS

The models were trained on historical data spanning from January 1, 2020, to December 31, 2023, and were subsequently evaluated using data from January 1, 2024, onwards. This approach ensures that the models are tested on unseen future data, providing a reliable assessment of their predictive capabilities in a real-world scenario. The model performance was evaluated using several key metrics, including Mean Absolute Error (MAE), Mean Squared Error (MSE), and the R² Score, allowing for a comprehensive analysis of prediction accuracy. These metrics provide insight into both the magnitude of prediction errors and the model's overall fit to the data. Rather than focusing on identifying peak demand times, the primary objective of this part is to first assess the basic prediction accuracy of the models.

A. Demand Forecast Precision

In terms of demand forecast precision, LightGBM emerged as the top-performing model. As shown in Table 2, it achieved the lowest MAE of 2.71×10^6 kW, indicating the smallest prediction errors and the most accurate forecasts of peak demand. LightGBM's performance is further supported by its lowest MSE of 13.77×10^{12} kW², reflecting fewer errors and a better fit to the demand data. Its highest R² Score of 0.8045 suggests that LightGBM effectively explains the variance in the demand data. Random Forest followed with an MAE of 3.03×10^6 kW and an MSE of 19.44×10^{12} kW². Although its MAE is higher compared to LightGBM, Random Forest still demonstrated strong performance with an R² Score of 0.7239, indicating reasonable accuracy and explanation of demand variance. Gradient Boosting, while not explicitly detailed here, generally performed similarly to Random Forest. KNN recorded a higher MAE of 3.21×10^6 kW and an MSE of 20.14×10^{12} kW², suggesting larger prediction errors compared to LightGBM and Random

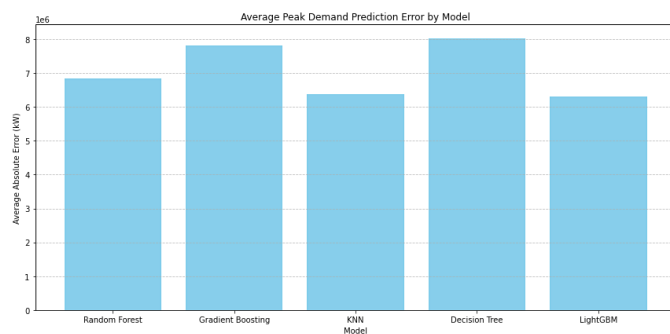


Figure.1. Average peak demand prediction error for each model

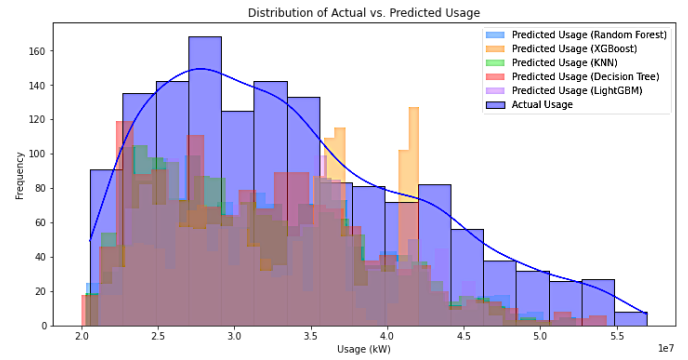


Fig. 2. Comparison of Predicted and Actual Electricity Demand with Multiple Machine Learning Models

Forest. Its R² Score of 0.7141 indicates lower explanatory power relative to LightGBM and Random Forest. Decision Tree had the highest MAE of 3.58×10^6 kW and the highest MSE of 28.21×10^{12} kW², reflecting the greatest prediction errors and the least effective model in forecasting peak demand. Its R² Score of 0.5994 underscores its relatively lower ability to explain the variance in demand data. The gap between actual and prediction as shown in Figure 1, LightGBM is still the best performing model.

Figure.2 presents a comparison between actual electricity usage and the predicted usage across five different machine learning models: Random Forest, XGBoost, KNN, Decision Tree, and LightGBM. The predicted usage values are represented as histograms with distinct colors for each model, while the actual usage is depicted as a smooth blue curve overlaying the histogram.

When analyzing the model predictions, Random Forest and LightGBM stand out as the models whose predictions most closely align with the actual usage curve, particularly in the mid-range of the data. This suggests that these models effectively capture the general trend of electricity consumption and are better suited for forecasting daily patterns. XGBoost and Decision Tree, on the other hand, display more scattered predictions with less consistency, particularly in capturing the peaks and troughs of actual usage. This inconsistency, especially in the higher usage ranges, suggests that these models may struggle to generalize across varying demand levels. KNN shows the most significant variance, especially at the lower and higher ends of the usage spectrum. This indicates that KNN, as a distance-based algorithm, may be overly sensitive to local fluctuations in the data, making it less reliable for extreme usage values.

Overall, the visual comparison highlights that LightGBM and Random Forest provide the closest match to actual usage patterns, with LightGBM performing slightly better in terms of accurately capturing the general shape of the demand curve. In contrast, models like XGBoost and KNN exhibit larger deviations from the actual usage, particularly in peak and offpeak periods, where accuracy is crucial for demand forecasting. This analysis suggests that for electricity demand forecasting, ensemble methods like LightGBM and Random Forest are better equipped to capture the underlying patterns in highvariance time series such as electricity consumption. Simpler models like KNN and Decision Tree, while capable of

offering insights, may struggle with the broader generalization required for such complex data, making them less effective for this specific forecasting task.

B. Peak Time Prediction Accuracy

As shown in Figure.3-5, the peak demand hours for the summer months between July and September exhibit distinct patterns. In 2023, the majority of the peaks occurred between 13:00 and 14:00, accounting for 76% of the total occurrences. This is a significant increase compared to previous years, where the corresponding peaks for the same period were 54% in 2022, 54% in 2021, and 41% in 2020. The growing concentration of peak times during these hours suggests an increasing reliance on cooling systems during these months, likely driven by higher temperatures and increased energy consumption for air conditioning.

This trend contrasts with winter months, where power demand peaks typically occur in two distinct periods, corresponding to the morning and evening hours due to heating needs. The variation in peak demand hours between the seasons requires different strategies for demand forecasting and management, ensuring that supply can meet demand efficiently throughout the year. In Figure.3, which presents the power demand forecast results for June 2, 2024, we perform daily demand forecasting and determine peak power based on these results. In this case, all models successfully captured the actual demand pattern and correctly identified the peak time. On this particular day, all models were able to predict the peak time within a one-hour margin.

However, only LightGBM was able to pinpoint the exact peak hour. In contrast, Figure.4 presents a case where all models missed the peak time by more than one hour. The actual peak occurred in the morning at 11:00, but all models incorrectly predicted a peak in the afternoon or evening. Furthermore, Figure.5 shows a day where only LightGBM correctly identified the actual peak at 13:00, while all other models failed to predict within the one-hour margin.

Figure.6 illustrates how accurately each model predicted the daily peaks for each month. From Figure.6, LightGBM also excelled in predicting peak times, with high accuracy rates in January (67.74%), March (77.42%), and July (74.19%). This demonstrates its superior ability to identify peak demand times accurately. Random Forest performed well in specific months, notably January (74.19%) and April (73.33%), though it showed less consistency than LightGBM. KNN and Decision Tree

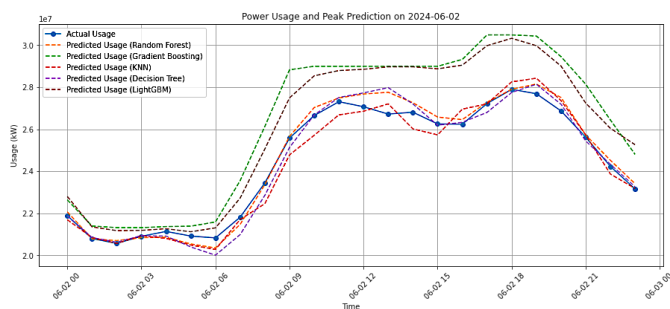


Fig. 3. Actual power usage and model predictions for peak demand on June 2, 2024, including peak times identified by each model.

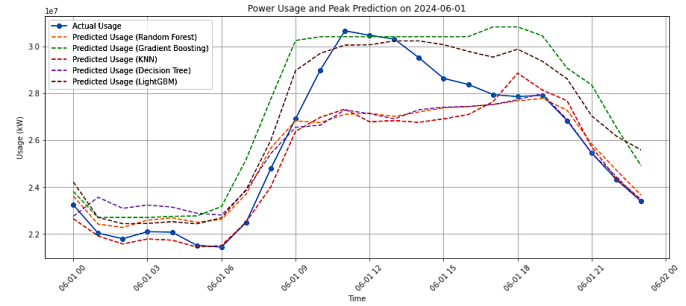


Fig. 4. Actual power usage and model predictions for peak demand on June 1, 2024, including peak times identified by each model.

exhibited variable accuracy, with lower overall performance in peak time prediction compared to LightGBM and Random Forest. XGBoost struggled the most with peak time accuracy, particularly in January (25.81%) and March (35.48%), indicating difficulties in accurate peak time forecasting.

C. Model Reliability and Suitability

The analysis indicates that LightGBM is the most reliable model for both predicting peak times and forecasting demand. Its combination of low MAE, low MSE, high R^2 Score, and high peak time prediction accuracy makes it particularly effective for strategies such as peak cutting and shifting. This is crucial for effective energy management and mitigating power shortages. Random Forest, while effective in certain scenarios, shows higher variability and a higher MAE compared to LightGBM. KNN and Decision Tree were less effective, with lower peak time prediction accuracy and higher MAE, making them less suitable for complex forecasting tasks. XGBoost had the lowest performance across the metrics, suggesting a need for further parameter tuning or data preprocessing.

In summary, LightGBM stands out as the optimal model for peak electricity forecasting. Its consistent performance across various metrics underscores its role as the best choice for enhancing energy management and preventing power shortages. The comparative analysis of models highlights the importance of selecting the right algorithm to achieve accurate and reliable forecasts in peak electricity demand. However, to further improve the accuracy of peak time predictions and ensure implementation in low-spec PC environments, the use of thresholds is also commonly employed. This will be discussed in more detail in the next Discussion section.

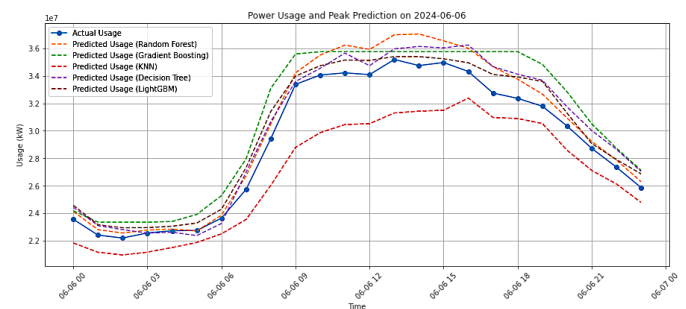


Fig. 5. Actual power usage and model predictions for peak demand on June 6, 2024, including peak times identified by each model.

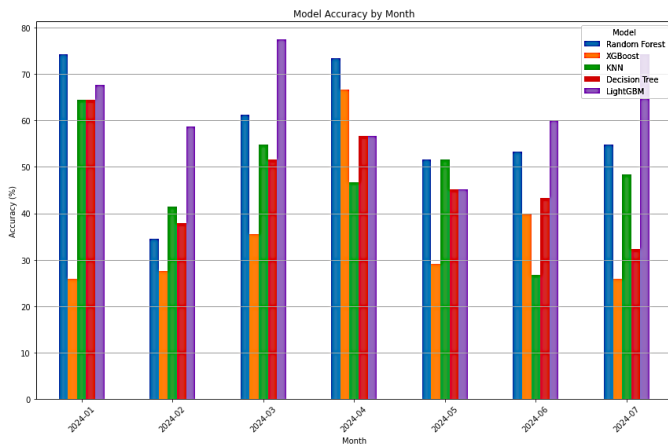


Fig. 6. Monthly accuracy percentages of different models for peak demand prediction during 2024.

IV. DISCUSSION

The capacity contribution fees are imposed on electricity retailers based on the total electricity usage during peak hours in each macro area per month. In other words, by accurately predicting the one-hour peak daily demand using a model like this, retailers can discharge batteries or request customers to conserve energy, minimizing the peak during that time. If such actions can be implemented, retailers can reduce the amount they need to pay compared to other competitors. Although the exact costs are not disclosed, it is estimated that several billion yen could be saved annually. Therefore, in addition to predicting the peak hour, decision-making thresholds for actions such as battery discharge or customer energy-saving requests would be highly beneficial for optimizing costs.

Figure.7 shows the predicted electricity demand for June 2024 and the one-hour peak within that month using the LightGBM model. As demonstrated, identifying the single peak hour out of 30 days, each day 24 hours in a month is not an easy task. However, one method to increase the probability of accurately predicting this peak is to use temperature thresholds. This approach involves setting thresholds based on temperature data, rather than directly incorporating the temperature data into the model, which does not require high PC specifications. Although this method still requires further research, a potential approach is to adjust the predicted values when the temperature exceeds the highest temperatures recorded over the past 3 to 5 years. For example, on June 24, 2024, which was an exceptionally hot day, such adjustments could improve prediction accuracy. Moving forward, improving this aspect of the model is necessary.

V. CONCLUSION

In summary, LightGBM has proven to be the most effective model for peak electricity forecasting, offering stable and

accurate performance that enhances energy management and reduces the risk of power shortages. Its superior accuracy makes it the optimal choice for implementing robust forecasting strategies, managing capacity contribution fees, and ensuring a reliable power supply. For electricity retailers, in particular, this can lead to significant reductions in the costs associated with capacity contribution fees, which can be seen as penalties, helping optimize energy supply and demand. While Random Forest provides value under certain conditions, its variability suggests it should be applied with caution. Models like KNN, Decision Tree, and XGBoost, which exhibit lower reliability and higher forecasting errors, are less suited for peak demand forecasting tasks. Additionally, in the Japanese market, there is a growing demand for explainable and relatively simple models, which can be a practical choice moving forward. These results underscore the importance of selecting the right model to achieve accurate forecasts and optimize energy management practices.

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